FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 3.3: WAVES	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

Your Final Explanation is your opportunity to show everything you have learned across all 8 lessons about waves. You are answering the big driving question: **How do waves carry energy through our environment — and how can understanding their behaviour protect Kenyan communities, improve technology, and help us make sense of the world around us?**

Read the driving question carefully. Your answer must connect back to the **anchoring phenomenon** — the fisherman on the shores of Lake Victoria near Kisumu who heard thunder bend around a rocky headland, watched water waves change direction at the shallow bay, and wondered why his mobile phone loses signal behind a hill in the Rift Valley.

Your task: Write a structured scientific explanation that uses evidence, wave behaviour vocabulary, and real Kenyan examples to answer the driving question fully. Work through each section below in order. Use your notes, your model drawings, and the Driving Question Board your class built together.

Before you begin, check that you can answer YES to each of these:

- Can I name and describe at least three wave behaviours (e.g., refraction, diffraction, reflection)?
- Can I explain what carries energy in a wave — and what does NOT move?
- Can I link each wave behaviour to a real situation in Kenya (weather, technology, safety, farming, or fishing)?
- Can I use at least one mathematical relationship (wave speed, frequency, wavelength) correctly?
- Can I explain how understanding waves helps protect or improve life for Kenyan communities?

Writing tips:

- Write in full sentences and organise your ideas clearly under each section heading.
- Use scientific vocabulary — but also explain what the words mean in plain language.
- Refer to the fisherman's observations, Lake Victoria, the Rift Valley, or other Kenyan contexts wherever possible.
- Aim for depth over length: one well-explained idea is better than five vague ones.

Section 1: What Is a Wave and How Does It Carry Energy?	
Start by explaining what a wave actually is. What moves — and what does NOT move? Describe the difference between transverse and longitudinal waves, and give one Kenyan example of each. Then explain the key wave properties: amplitude, wavelength, frequency, and wave speed. Use the wave speed equation ($v = f\lambda$) in at least one example. Connect your answer back to the fisherman: when he hears the thunder rumble, what kind of wave is travelling through the air to reach his ears?	A wave is a disturbance that transfers energy from one place to another without permanently moving the matter it travels through. When the fisherman hears thunder across Lake Victoria, sound energy travels to his ears as a longitudinal wave — a series of compressions and rarefactions moving through the air. The air particles vibrate back and forth along the direction the sound travels, but no air particle actually travels from the lightning strike to the fisherman's ear. This is the key idea: energy moves, matter oscillates in place. By contrast, the water waves spreading outward from the storm's centre are transverse waves — the water surface moves up and down (perpendicular to the direction the wave travels), while the wave energy radiates outward across the lake. You can picture this when you watch a fisherman's buoy bobbing up and down as a wave passes: the buoy rises and falls but does not travel toward the shore with the wave. Every wave has four measurable properties. Amplitude is the maximum displacement from the rest position — for a sound wave, a larger amplitude means a louder thunder crack. Wavelength (λ) is the distance between two successive crests (or compressions). Frequency (f) is the number of complete waves passing a point per second, measured in hertz (Hz). Wave speed (v) links these: $v = f\lambda$. For example, if a sound wave in air has a frequency of 170 Hz and a wavelength of 2 m, its speed is $v = 170 \times 2 = 340$ m/s — roughly the speed of sound in warm East African air. This is why the fisherman sees the lightning flash before he hears the thunder: light travels far faster (about 3×10^8 m/s) than sound, so the light arrives almost instantly while sound takes several seconds to cover the same distance.

Section 2: Reflection and Refraction — Why Waves Change Direction	
The fisherman noticed that water waves changed direction as they moved from the deep open lake into the shallow bay near the shore. Explain what refraction is, why it happens, and what changes (and what stays the same) when a wave refracts. Then explain reflection — bouncing back from a surface. Use Snell's Law or a ray diagram description to support your explanation. Give at least one example from Kenya for each behaviour (for example: sonar used by fishing vessels on Lake Victoria, or the echo of sound off the cliffs at Hell's Gate in Naivasha). How do these two behaviours help explain what the fisherman observed?	When the water waves travelling across the deep open lake reached the shallower water of the bay near Kisumu, they slowed down. In deep water, waves travel faster; in shallow water, friction with the lake bed slows them. Because one side of an incoming wave front enters the shallower zone before the other side, that side slows first — causing the whole wave front to bend toward the shallower region. This bending is called refraction. Crucially, the frequency of the wave does not change during refraction (the same number of wave crests pass any point per second), but the wavelength shortens because $v = f\lambda$ and v has decreased. This is why the fisherman sees the waves curve and seem to 'aim' at the shallow shore rather than continuing in a straight line. Reflection happens when a wave bounces off a surface. The angle at which the wave hits the surface (the angle of incidence) equals the angle at which it bounces away (the angle of reflection). Fishing vessels on Lake Victoria use sonar — they send out pulses of sound waves downward into the water; those waves reflect off the lake bed (or off schools of fish) and return to the vessel. By measuring the time taken for the echo to return and knowing the speed of sound in water (~1 500 m/s), the crew can calculate the depth: depth = $(v \times t) \div 2$. This technology helps fishermen like those operating out of Kisumu port to locate tilapia and Nile perch more efficiently. Together, refraction and reflection explain the fisherman's first puzzle: the water waves he observed bending into the bay were refracting as they crossed from deep to shallow water — not travelling in a new straight line, but curving because different parts of the wave front slowed at different moments.

Section 3: Diffraction — How Waves Spread Around Obstacles	
The fisherman could hear the thunder even though the storm was directly behind a rocky headland — there was no straight-line path for the sound. Explain what diffraction is and when it is most significant (hint: think about the relationship between wavelength and the size of the gap or obstacle). Use a diagram description or a drawn diagram to support your explanation. Then connect diffraction to the mobile phone signal problem in the Rift Valley: why does a signal fade behind a hill, and why does it not disappear completely? Give one more Kenyan example where diffraction is either useful or a problem.	Diffraction is the spreading of a wave as it passes through a gap or around the edge of an obstacle. It is the reason the fisherman hears thunder even when the rocky headland blocks the straight-line path between him and the storm. Sound waves have relatively long wavelengths (roughly 0.02 m to 17 m for the range of human hearing), so when they encounter the headland — whose width is comparable to or smaller than those wavelengths — they spread significantly around the edges, bending into the 'shadow zone' behind the cliff. Diffraction is most pronounced when the wavelength of the wave is similar to, or larger than, the size of the obstacle or gap. This is why you can hear someone calling you from around the corner of a building (sound diffracts easily around the corner) but you cannot see them (light has a wavelength of roughly 400–700 nm, far smaller than a building, so it barely diffracts). In the Rift Valley, mobile phone signals travel as radio waves. When you walk behind a hill, the hill blocks the straight-line path to the transmission tower. However, radio waves (with wavelengths of centimetres to metres) diffract around and over the hill to some extent — which is why your signal weakens but does not instantly drop to zero. Engineers at Safaricom and other providers use this understanding to position relay towers along the Rift Valley escarpment so that diffracted signals can still reach communities in valleys below. Another Kenyan example: the entrance channels to Kilindini Harbour in Mombasa are engineered so that ocean wave energy is partially diffracted into the harbour basin in a controlled way, reducing the destructive force of large swells on moored vessels — an application of diffraction in coastal engineering.

Section 4: The Electromagnetic Spectrum and Wave Applications in Kenya	
Not all waves need matter to travel through — electromagnetic waves can move through a vacuum. List the regions of the electromagnetic spectrum in order of increasing frequency (or decreasing wavelength). Then choose THREE types of electromagnetic waves and explain one specific, beneficial application of each in a Kenyan context (for example: how radio waves support community FM stations in rural areas, how infrared is used in thermal imaging for wildlife conservation in Maasai Mara, or	Electromagnetic (EM) waves are transverse waves that carry energy through electric and magnetic field oscillations. Unlike sound or water waves, they do not need a medium — they travel through the vacuum of space at the speed of light (3×10^8 m/s). The EM spectrum, arranged from lowest to highest frequency (and longest to shortest wavelength), is: radio waves → microwaves → infrared → visible light → ultraviolet → X-rays → gamma rays. As frequency increases, so does the energy carried by each photon. **Radio waves** have very long wavelengths (from 1 mm to over 100 km) and low energy, which means they can travel long distances and diffract around hills and buildings. Community radio stations such as Kameme FM and Radio Lake Victoria use radio waves to broadcast news, health information, and weather warnings to rural farming and fishing communities across Kenya who may not have reliable internet access. Their long wavelength is what allows the signal to diffract over the Aberdare Range and reach listeners in remote valleys. **Infrared radiation** has a wavelength just longer than visible red light. All warm objects emit infrared, and the hotter the object, the more intensely it radiates. Kenya Wildlife Service rangers and researchers in Maasai Mara use infrared thermal cameras to track elephant and lion movements at night without disturbing the animals or needing artificial light. The cameras detect the infrared energy radiated by the animals' warm bodies, producing images based on temperature differences — protecting both wildlife and local Maasai communities.

how UV radiation affects tea farmers in Kericho). For each application, explain which wave property (frequency, wavelength, or energy) makes that type of wave suitable for that use.	**Ultraviolet (UV) radiation** has a higher frequency and more energy than visible light. Tea farmers in Kericho must understand UV exposure because excessive UV damages the DNA in plant cells, reducing crop yield — which is why tea bushes are often grown under partial shade. At the same time, public health workers use UV lamps in rural health centres to sterilise surfaces and water, because UV energy at specific frequencies is absorbed by bacterial DNA, preventing reproduction. Understanding the dual nature of UV — harmful at high doses, useful when controlled — is a direct application of wave energy knowledge.

Section 5: Pulling It All Together — How Wave Knowledge Protects and Improves Life in Kenya	
Now answer the driving question directly and fully. Looking back at the fisherman's observations on Lake Victoria, explain ALL three of his puzzles using correct scientific language: (1) why he hears thunder after the lightning flash, (2) why sound bends around the headland, and (3) why water waves change direction at the shallower bay. Then zoom out: write two or three paragraphs explaining how understanding wave behaviour — including reflection, refraction, diffraction, and the electromagnetic spectrum — can protect Kenyan communities from natural hazards, improve everyday technology, and help people make better decisions. Use at least two specific Kenyan examples that you have not already used in earlier sections.	The fisherman's three puzzles all have wave science at their heart. First, he sees the lightning before he hears the thunder because light and sound are different kinds of waves travelling at very different speeds. Light is an electromagnetic wave travelling at 3×10^8 m/s — it crosses even 10 kilometres almost instantly. Sound is a mechanical longitudinal wave that travels at roughly 340 m/s in warm air. For every 3 seconds between the flash and the rumble, the storm is approximately 1 km away. This is why Kenyan farmers and fishermen are taught the 'flash-to-bang' rule as a lightning safety measure. Second, sound bends around the rocky headland because of diffraction. The long wavelengths of low-frequency thunder (some wavelengths exceed the width of the headland) allow the sound wave to spread significantly around the edges of the obstacle, filling in the geometric 'shadow' behind the cliff. Higher-frequency sounds with shorter wavelengths would diffract less, which is why the deep rumble of thunder reaches the fisherman more easily than a sharp, high-pitched crack would. Third, the water waves change direction as they cross from deep to shallow water because of refraction. The wave front slows on the shallow side first, causing the whole front to pivot — bending toward the shallower region. The frequency stays constant, but the wavelength and speed both decrease. This explains why waves on Kenyan beaches always seem to arrive nearly parallel to the shoreline regardless of the direction of the wind: refraction continuously bends the wave fronts to align with the underwater contour of the shore. Zooming out, wave science has life-saving and life-improving applications across Kenya. The Kenya Meteorological Department uses weather radar — which sends out microwave pulses and detects the reflected signals from rain droplets — to issue flood warnings for communities along the Tana River and Nyando Basin, where seasonal flooding can be catastrophic. By understanding how microwaves reflect off precipitation (a direct application of radar reflection physics), forecasters can track storm cells and give farmers and families advance warning to move livestock and secure homes. In education and communication, understanding wave behaviour helps explain why fibre-optic internet cables — which carry data as pulses of visible light using total internal reflection inside glass fibres — can transmit information across Kenya's National Optic Fibre Backbone Infrastructure (NOFBI) almost instantaneously, connecting schools in Turkana and Mandera to online resources. Every time a student in a remote area loads a lesson on a tablet, they are benefiting from the principle that light waves can be 'trapped' inside a medium by reflection at the boundary — and guided across hundreds of kilometres without significant energy loss. Ultimately, the fisherman's puzzled observations on the shores of Lake Victoria are not isolated curiosities. They are windows into

	a set of universal wave behaviours — reflection, refraction, diffraction, and energy transfer — that underpin weather forecasting, telecommunications, medical imaging, coastal engineering, and environmental protection. Understanding waves does not just explain puzzles; it gives Kenyan communities the scientific tools to build safer infrastructure, respond to natural hazards, and innovate for the future.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific Accuracy of Wave Concepts	All wave concepts (transverse vs. longitudinal, reflection, refraction, diffraction, EM spectrum) are defined and applied correctly throughout. The wave speed equation ($v = f\lambda$) is used accurately with at least one worked example. No misconceptions are present.	Most wave concepts are correctly defined and applied. The wave speed equation is mentioned and mostly used correctly, with minor errors. One or two small inaccuracies are present but do not undermine the overall explanation.	Some wave concepts are mentioned but several are confused, incomplete, or incorrect (e.g., confusing transverse and longitudinal, or misapplying refraction and diffraction). The wave speed equation is absent or incorrectly used.
Explanation of the Anchoring Phenomenon	All three of the fisherman's observations (time delay between lightning and thunder, sound bending around the headland, water waves changing direction) are fully and correctly explained using precise scientific language. The explanations are clearly connected to the relevant wave behaviour (speed difference, diffraction, refraction).	At least two of the fisherman's three observations are correctly explained with scientific reasoning. The third is partially addressed or contains a minor error. Connections to wave behaviours are mostly clear.	Only one observation is explained, or the explanations are vague and rely on everyday language without scientific reasoning (e.g., 'the sound bounces around' without explaining diffraction). The phenomenon is not clearly linked to specific wave behaviours.
Use of Kenyan Contexts and Real-World Applications	At least four distinct, accurate Kenyan examples are used (e.g., Lake Victoria fishing, Rift Valley mobile signals, Kericho tea farming, Mombasa harbour, Kenya Meteorological Department radar, NOFBI fibre optics). Each example is correctly linked to a specific wave behaviour or property.	Two or three Kenyan examples are used and mostly correctly linked to wave behaviour. Examples are relevant and show an attempt to connect science to local real-world contexts.	Fewer than two Kenyan examples are used, or examples are generic (e.g., 'mobile phones use radio waves') without specific Kenyan detail or a clear link to wave science concepts.
Depth of Reasoning and Use of Evidence	Explanations consistently go beyond naming concepts — the student explains WHY each wave behaviour occurs (e.g., explains that refraction happens because wave speed changes at the boundary, not just that 'waves bend'). Scientific reasoning is logical, sequential, and supported by evidence from lessons and investigations.	Most explanations go beyond naming to provide some reasoning. A few explanations remain at the surface level ('waves refract in shallow water') without explaining the underlying cause. Reasoning is generally logical.	Explanations mostly name concepts without reasoning. The student states what happens but rarely explains why. Little or no reference is made to evidence gathered in lessons or investigations.

Communication, Structure, and Use of Scientific Vocabulary	The explanation is clearly organised with all five sections addressed. Scientific vocabulary (amplitude, wavelength, frequency, refraction, diffraction, reflection, transverse, longitudinal, electromagnetic spectrum) is used accurately and consistently throughout. Writing is clear, precise, and accessible.	Most sections are addressed and the explanation is reasonably well organised. Scientific vocabulary is used in most sections, with occasional imprecision. The writing is mostly clear with some lapses.	The explanation is poorly organised or several sections are missing. Scientific vocabulary is rarely used or frequently misused. The writing is difficult to follow and relies mostly on everyday language without precision.